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ORIGINAL RESEARCH

Clinical Outcomes and Repeat-CRFA-Free Survival After Ultrasound-Guided Cooled Radiofrequency Ablation Using the Matsuoka Direct Nerve Targeting (MDNT) Technique for Knee Osteoarthritis: A Retrospective Cohort Study of 1,025 Consecutive Knees

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Background: Cooled radiofrequency ablation (CRFA) is a minimally invasive treatment for knee osteoarthritis (KOA) pain. Conventional techniques commonly use anatomical landmarks to estimate genicular nerve location. The Matsuoka Direct Nerve Targeting (MDNT) Technique combines ultrasound visualization with physiological confirmation before ablation. We evaluated clinical outcomes and repeat-CRFA-free survival after CRFA using this technique.

Methods: This retrospective single-center cohort included 1,025 consecutive knees with symptomatic KOA treated between June 2024 and June 2026. NRS pain scores were recorded preoperatively, on postoperative day 1, and at approximately 1 year when available through rolling telephone follow-up. Repeat CRFA, complications, and conversion to total knee arthroplasty were secondary outcomes. Repeat-CRFA-free survival was estimated using Kaplan-Meier analysis.

Results: Among 1,021 knees with complete early NRS data, mean NRS decreased from 7.76 ± 1.31 preoperatively to 2.46 ± 1.93 on postoperative day 1 (mean reduction, 5.29; 95% CI, 5.16–5.43; $P < 0.001$). One-year NRS data were available for 246 knees, in which mean NRS decreased from 8.15 ± 1.12 preoperatively to 2.29 ± 2.06 at approximately 1 year. Repeat CRFA was performed in 8 of 1,025 knees (0.8%; 95% CI, 0.4–1.5%), and TKA was reported as planned in 1 knee (0.1%). Kaplan-Meier repeat-CRFA-free survival was 99.8% at 12 months and 96.9% at 20 months. No major procedure-related complications were observed.

Conclusion: Ultrasound-guided CRFA using MDNT was associated with substantial early pain reduction, sustained pain relief among knees with available one-year follow-up, a low observed repeat-CRFA rate, and a favorable safety profile. Because this was a retrospective single-center, single-operator study without a comparator, the findings are hypothesis-generating and do not establish superiority over established techniques. Prospective multicenter comparative studies are warranted.

Keywords: knee osteoarthritis, cooled radiofrequency ablation, genicular nerve, ultrasound-guided intervention, direct nerve targeting, Kaplan-Meier analysis, repeat-CRFA-free survival

Introduction

Knee osteoarthritis (KOA) is one of the most common musculoskeletal disorders worldwide and a leading cause of chronic pain and disability among older adults. As life expectancy continues to increase, the global prevalence of KOA is expected to rise substantially, placing an increasing burden on healthcare systems and society. Persistent knee pain

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negatively affects walking ability, activities of daily living, physical function, and health-related quality of life, making effective pain management an important therapeutic goal.[1,2]

Current treatment strategies for KOA include patient education, weight reduction, exercise therapy, oral analgesics, non-steroidal anti-inflammatory drugs (NSAIDs), and intra-articular injections. Although these conservative treatments provide symptomatic relief for many patients, a considerable proportion continue to experience persistent pain despite optimized management. International clinical practice guidelines recommend individualized treatment according to disease severity and patient characteristics, while recognizing the need for additional minimally invasive treatment options before total knee arthroplasty (TKA).[3–7]

TKA remains the standard surgical treatment for end-stage KOA; however, many patients are not suitable candidates because of advanced age, frailty, multiple medical comorbidities, anticoagulant therapy, or a preference to avoid surgery. Furthermore, postoperative persistent pain remains a recognized clinical problem in a subset of patients undergoing TKA. Consequently, minimally invasive interventions capable of providing durable pain relief have attracted increasing attention.[2,8]

Among these interventions, cooled radiofrequency ablation (CRFA) of the genicular sensory nerves has emerged as an effective treatment for chronic knee pain associated with KOA. Several prospective randomized controlled trials have demonstrated that CRFA provides superior and longer-lasting pain relief than intra-articular corticosteroid or hyaluronic acid injections, with clinically meaningful improvements in pain and function extending for at least 12 months.[9–12]

Despite these favorable clinical outcomes, established CRFA techniques commonly localize target nerves using osseous landmarks, adjacent vascular structures, or predefined anatomical target zones. Ultrasound-guided genicular nerve procedures have also been described and can improve anatomical visualization. However, the course of sensory nerves varies among individuals, and an expected anatomical target does not necessarily confirm that the symptomatic sensory nerve itself has been identified.

The MDNT Technique was developed as a technical modification of ultrasound-guided CRFA rather than as a claim of being the first ultrasound-guided genicular nerve ablation method. Its proposed distinction is the standardized integration of direct sonographic visualization of a nerve-like target at the symptomatic site with physiological confirmation using concordant pain reproduction, sensory stimulation, and the characteristic response after initiation of radiofrequency energy before definitive ablation. This approach is intended to increase confidence in target localization, but whether it improves clinical outcomes over established targeting techniques remains unproven.

The purpose of this study was to describe clinical outcomes after ultrasound-guided CRFA performed using the MDNT Technique in a consecutive cohort of 1,025 knees with symptomatic KOA. We assessed early pain outcomes, available one-year pain outcomes, repeat CRFA, procedure-related complications, and repeat-CRFA-free survival using Kaplan-Meier analysis. The study was not designed to establish superiority over conventional targeting methods.

Methods

Study Design and Ethical Approval

This retrospective single-center cohort study evaluated consecutive knees with symptomatic knee osteoarthritis (KOA) treated using ultrasound-guided cooled radiofrequency ablation (CRFA) with the Matsuoka Direct Nerve Targeting (MDNT) Technique between June 4, 2024, and June 30, 2026.

The study was conducted in accordance with the principles of the Declaration of Helsinki and is reported following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement.[13]

This retrospective observational study was conducted at Matsuoka Emergency Clinic, Minamikyushu, Kagoshima, Japan.

The study protocol was approved by the Ethics Committee of Wide Area Medical Corporation EMS (Approval No. 2601; approved on June 1, 2024) before the initiation of the study.

Patients

The unit of analysis was the treated knee.

A total of 1,025 consecutive knees undergoing CRFA during the study period were included.

Inclusion Criteria

Patients were eligible if they met all of the following criteria:

- Radiographically diagnosed symptomatic KOA
- Persistent knee pain despite conservative treatment
- Inadequate symptom relief following medication and/or intra-articular injections
- Willingness to undergo CRFA after discussion of available treatment options

Exclusion Criteria

Patients were excluded if they had

- Septic arthritis
- Active local infection
- Malignant tumor involving the knee
- Severe cognitive impairment
- Clinically significant coagulation disorders
- Inability to complete postoperative follow-up

Demographic characteristics including age, sex, laterality, and Kellgren–Lawrence (KL) grade were recorded.

MDNT Technique

All procedures were performed by the senior author (Y.M.) using the Coolief® Cooled Radiofrequency System (Avanos Medical, Alpharetta, GA, USA) under ultrasound guidance.

Ultrasound imaging was performed using a Canon Aplio i800 ultrasound system (Canon Medical Systems Corporation, Tochigi, Japan) equipped with an 18-MHz linear-array transducer.

All procedures were performed by the senior author (Y.M.), who developed the MDNT Technique and has more than 20 years of clinical experience in musculoskeletal interventions. This standardized single-operator approach reduced procedural heterogeneity within the cohort but also limits assessment of inter-operator reproducibility and may introduce operator- and developer-related bias.

Unlike conventional CRFA techniques that estimate the target nerve using osseous landmarks or adjacent vessels, the MDNT Technique aims to identify the responsible sensory nerve directly before ablation.

The procedure consisted of the following sequential steps.

Step 1. Identification of the painful point

The point of maximal tenderness was identified by manual palpation before ultrasonographic examination.

Step 2. Ultrasound identification of the target nerve

High-resolution ultrasonography was used to identify a hypoechoic nerve-like structure immediately superficial to the periosteum corresponding to the painful region.

Step 3. Electrode placement

The radiofrequency electrode was advanced toward the center of the visualized sensory nerve while maintaining gentle contact with the periosteum.

Step 4. Physiological confirmation

Correct localization of the target nerve required confirmation by all of the following findings:

1. Direct ultrasound visualization of the target nerve
2. Reproduction of the patient's characteristic pain when the electrode contacted the nerve

3. Concordant sensory stimulation
4. Characteristic pain immediately after initiation of radiofrequency energy followed by gradual attenuation

Definitive ablation was performed only after all confirmation criteria supported accurate localization of the responsible sensory nerve.

For knee osteoarthritis, three genicular sensory nerve targets were routinely treated.

Radiofrequency energy was delivered at 60°C for 150 seconds (2.5 minutes) at each treatment site.

Postoperative Management

Patients returned to the outpatient clinic on postoperative day 1 for wound assessment and evaluation of pain using the Numerical Rating Scale (NRS).

No routine restriction on weight-bearing or daily activities was imposed.

Analgesic medication was prescribed only when required for breakthrough pain.

Routine postoperative rehabilitation was not recommended; however, patients requesting rehabilitation were referred individually according to clinical need.

Outcome Measures

The primary outcome was pain intensity measured using the Numerical Rating Scale (NRS; 0–10).

NRS scores were recorded

- Before treatment
- On postoperative day 1
- Approximately one year after treatment

One-year NRS outcomes were obtained by standardized telephone follow-up conducted by trained clinic staff using a standardized questionnaire. Because enrollment was continuous through June 2026, one-year NRS data were not expected to be available for the full cohort at the time of manuscript preparation. Telephone follow-up was initiated on a rolling basis after each treated knee reached approximately one year. All successfully contacted eligible knees were included in the one-year analysis. Some patients who had reached the one-year time point could not be contacted despite attempted telephone follow-up.

Secondary Outcomes Included

- Repeat CRFA
- Repeat-CRFA-free survival
- Conversion to total knee arthroplasty
- Procedure-related complications

Procedure-related complications included infection, hematoma, skin burn, neurological deficit, vascular injury, and any adverse event requiring additional treatment.

All treated knees were followed for repeat CRFA until June 30, 2026.

Statistical Analysis

Continuous variables are presented as mean ± standard deviation (SD), whereas categorical variables are presented as frequency and percentage.

Paired changes in NRS were analyzed using the paired Student's *t*-test.

Mean paired differences are reported with 95% confidence intervals (CIs).

Effect size was calculated using Cohen's dz.[14]

The crude repeat intervention rate was calculated together with Wilson 95% confidence intervals. No a priori sample-size or power calculation was performed because all consecutive eligible knees treated during the predefined study period were included.

Repeat-CRFA-free survival was estimated using the Kaplan-Meier method, with repeat CRFA defined as the event of interest and the last available follow-up treated as censoring.[15] The treated knee was the unit of analysis. For bilateral knees, within-patient correlation was not modeled; therefore, knee-level confidence intervals and inferential statistics should be interpreted with this limitation in mind.

Statistical analyses were performed using Python version 3.11 with the SciPy and statsmodels libraries.

A two-sided P value of <0.05 was considered statistically significant.

Results

Patient Characteristics

Between June 4, 2024, and June 30, 2026, a total of 1,025 consecutive knees underwent ultrasound-guided cooled radiofrequency ablation (CRFA) using the Matsuoka Direct Nerve Targeting (MDNT) Technique.

Complete preoperative and postoperative day-1 Numerical Rating Scale (NRS) data were available for 1,021 knees (99.6%), which constituted the primary early pain cohort. Four knees had incomplete NRS documentation and were excluded from the early pain analysis but remained in the repeat-CRFA follow-up cohort. The mean patient age was 76.3 ± 5.7 years. The knee-level demographic data included 449 male and 576 female knees, with 524 right and 501 left knees treated.

Radiographic severity according to the Kellgren–Lawrence (KL) classification consisted of 11 knees (1.1%) with grade II, 727 knees (70.9%) with grade III, and 287 knees (28.0%) with grade IV.

Repeat CRFA was performed in 8 of 1,025 knees, corresponding to a crude repeat intervention rate of 0.8% (95% Wilson confidence interval [CI], 0.4–1.5%).

Baseline demographic and clinical characteristics are summarized in Table 1.

The patient flow and availability of outcome data are summarized in Figure 1.

Pain Outcomes

Substantial pain reduction was observed immediately following treatment.

Among the 1,021 knees included in the primary efficacy analysis, mean NRS significantly decreased from 7.76 ± 1.31 preoperatively to 2.46 ± 1.93 on postoperative day 1, corresponding to a mean reduction of 5.29 points (95% CI, 5.16–5.43; $P < 0.001$). The effect size was very large (Cohen's dz = 2.40).

Table 1 Baseline Characteristics

Characteristic	Value
Total treated knees	1,025
Age, years (mean ± SD)	76.3 ± 5.7
Sex	Male 449 (43.8%); Female 576 (56.2%)
Side	Right 524 (51.1%); Left 501 (48.9%)
KL grade II	11 (1.1%)
KL grade III	727 (70.9%)
KL grade IV	287 (28.0%)
Repeat CRFA	8/1,025 (0.8%; 95% CI, 0.4–1.5%)
TKA recommended/scheduled at follow-up	1/1,025 (0.1%)

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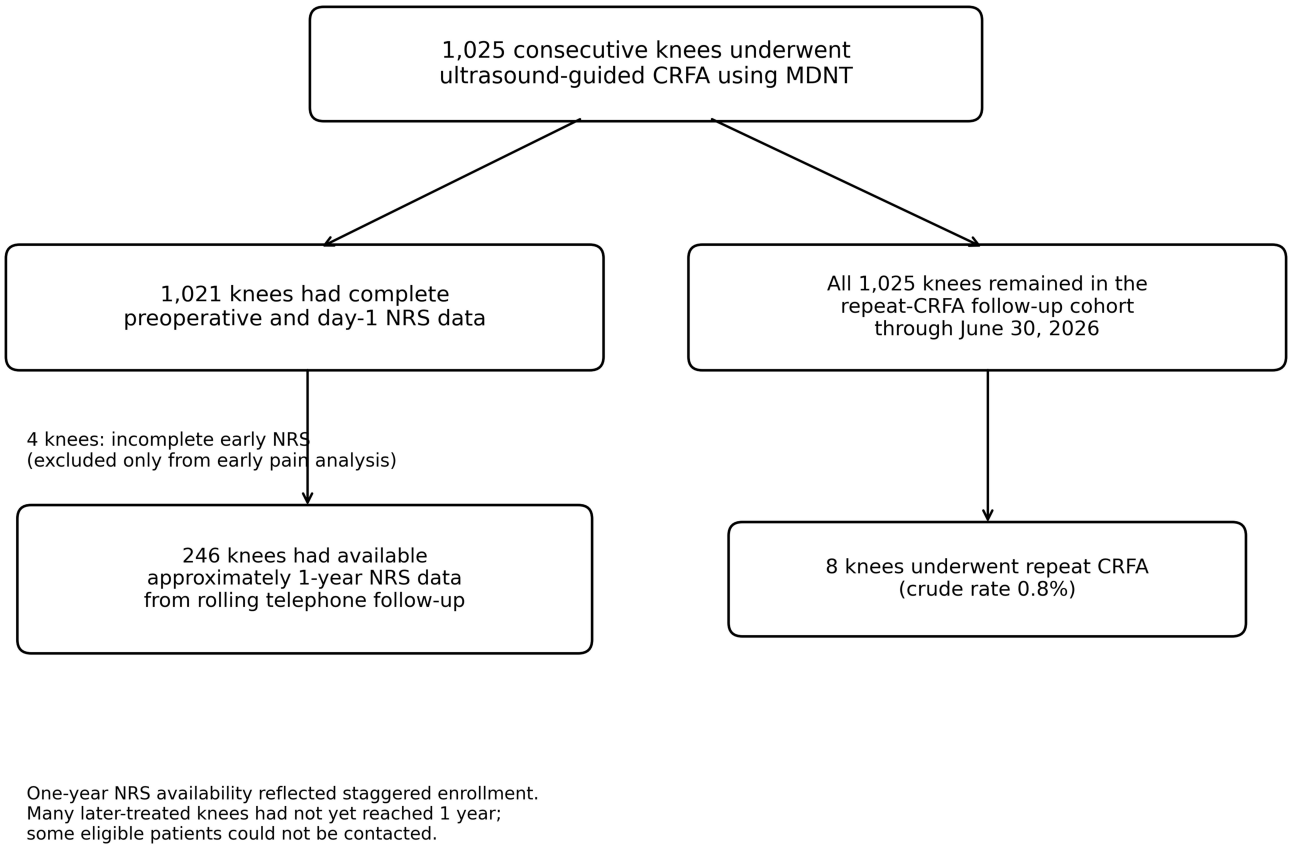


Figure 1 Patient flow and availability of outcome data. Four knees with incomplete early NRS documentation were excluded only from the early pain analysis and remained in the repeat-CRFA follow-up cohort. One-year NRS assessments were obtained by rolling telephone follow-up after approximately one year; many later-treated knees had not yet reached the one-year time point at manuscript preparation, and some eligible patients could not be contacted.

At the time of manuscript preparation, standardized one-year telephone NRS assessments had been completed for 246 knees. This was an available-case cohort generated by rolling follow-up rather than a prespecified subset: many later-treated knees had not yet reached the one-year time point, while some eligible patients could not be reached by telephone. All successfully contacted eligible knees were included. In these 246 knees, mean NRS improved from 8.15 ± 1.12 preoperatively to 2.12 ± 1.98 on postoperative day 1 and remained low at 2.29 ± 2.06 at approximately one year.

A simple baseline comparison of the 246 knees with one-year NRS data versus all remaining knees was not performed because the remaining group combined administratively immature observations (knees that had not yet reached one year) with a smaller subset of eligible but unreachable patients; these represent different missing-data mechanisms and do not constitute a single conventional lost-to-follow-up group.

The mean reduction from baseline to one year was 5.85 points (95% CI, 5.55–6.16; $P < 0.001$), corresponding to a very large effect size (Cohen's dz = 2.44).

Mean NRS increased slightly between postoperative day 1 and the one-year follow-up by 0.17 points.

Pain outcomes are summarized in Table 2. Pain trajectories are illustrated in Figures 2 and 3, demonstrating substantial early pain reduction that was maintained throughout the available follow-up period. Error bars indicate 95% confidence intervals.

Table 2 Pain Outcomes

Outcome	n	Preoperative	POD1	1 Year	Mean Reduction	95% CI	P value	Cohen's dz
Early outcome	1,021	7.76 ± 1.31	2.46 ± 1.93	—	5.29	5.16–5.43	<0.001	2.40
One-year follow-up	246	8.15 ± 1.12	2.12 ± 1.98	2.29 ± 2.06	5.85	5.55–6.16	<0.001	2.44

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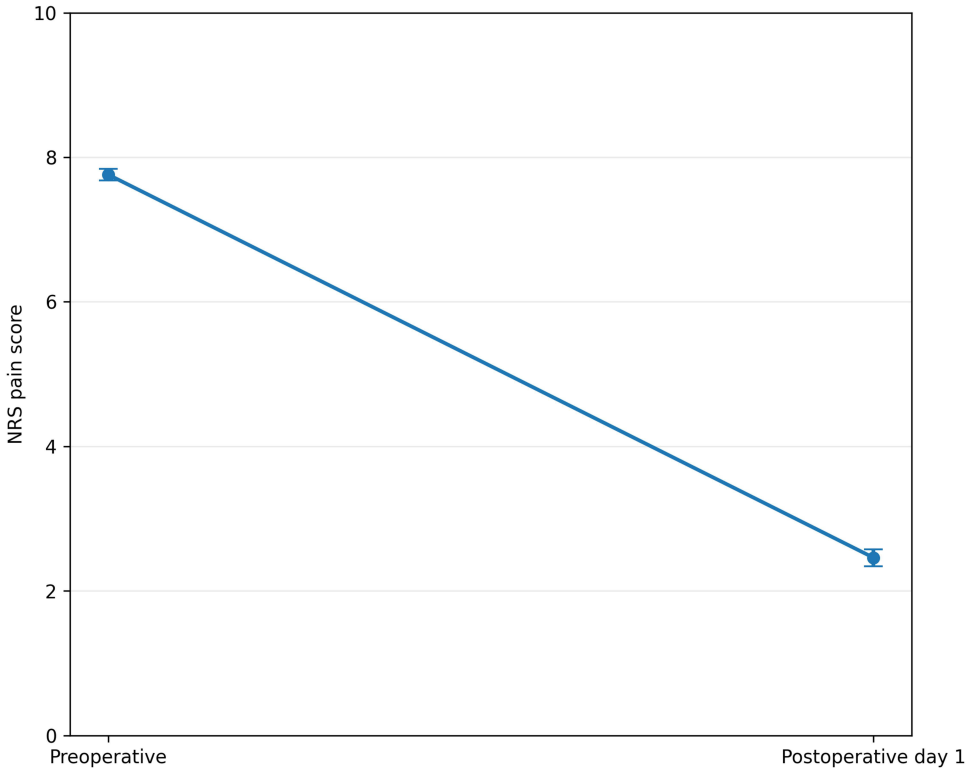


Figure 2 Early NRS improvement from preoperative assessment to postoperative day 1 in 1,021 knees with complete early outcome data. Error bars indicate 95% confidence intervals.

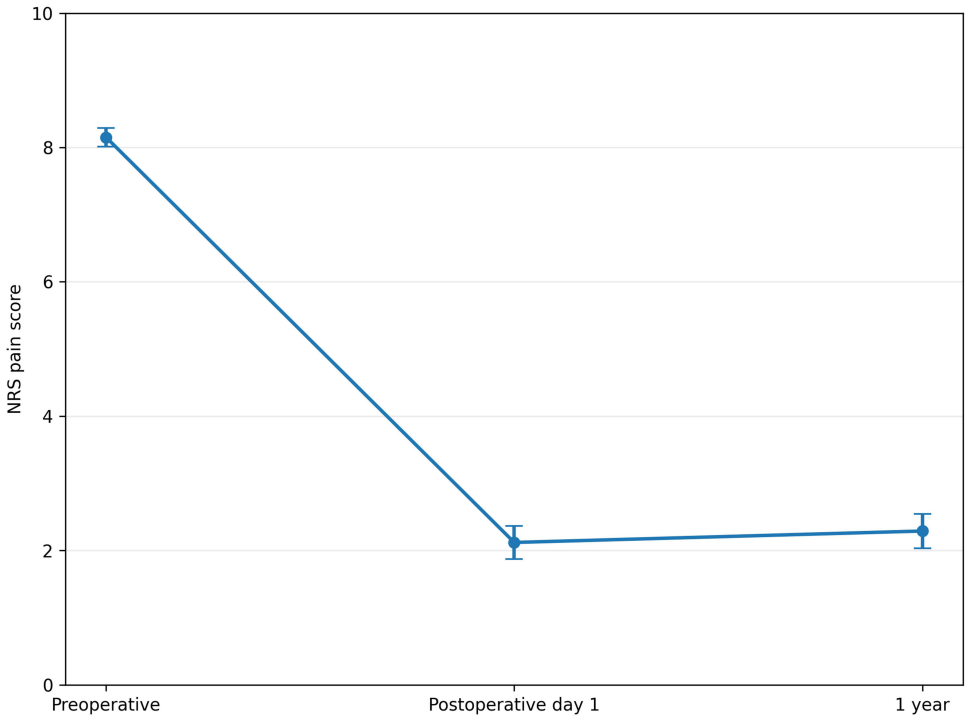


Figure 3 NRS trajectory in the 246 knees with available approximately one-year telephone follow-up data. Error bars indicate 95% confidence intervals.

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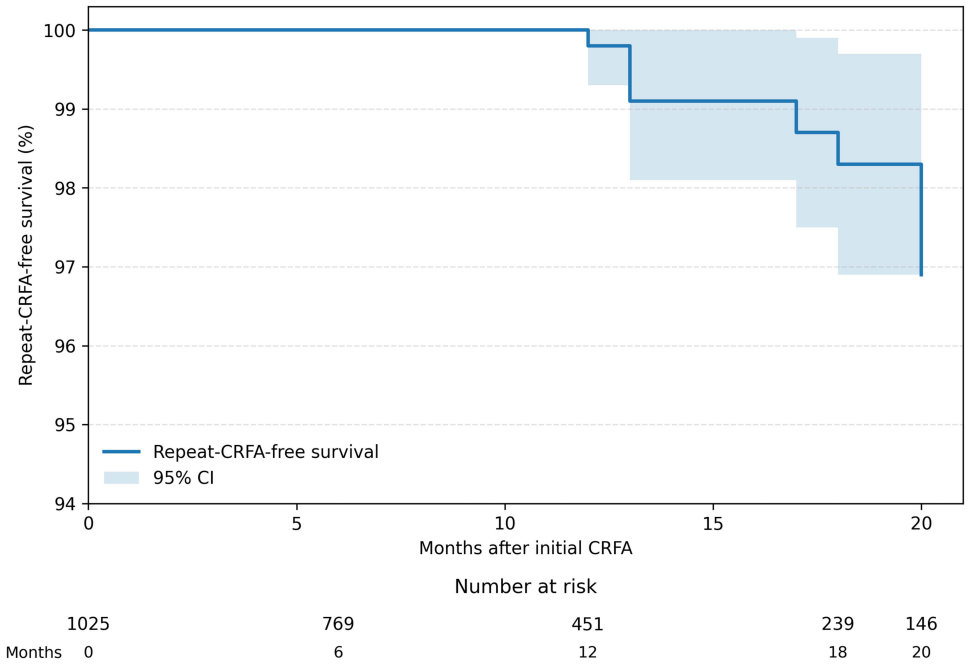


Figure 4 Kaplan-Meier curve for repeat-CRFA-free survival after ultrasound-guided CRFA using the MDNT Technique. Repeat CRFA was defined as the event of interest. The shaded region represents the 95% confidence interval, and numbers at risk are shown below the x-axis. Repeat-CRFA-free survival should not be interpreted as objective treatment-failure-free survival because repeat intervention is a clinical treatment decision.

Repeat-CRFA-Free Survival

Repeat CRFA was required in 8 of 1,025 knees, yielding a crude repeat intervention rate of 0.8% (95% CI, 0.4–1.5%).

Repeat procedures occurred between 12 and 20 months after the initial treatment.

Kaplan–Meier analysis demonstrated sustained Repeat-CRFA-free survival throughout the observation period, with estimated Repeat-CRFA-free survival rates of 99.8% at 12 months, 98.3% at 18 months, and 96.9% at 20 months.

The crude proportion without repeat CRFA at the end of follow-up was 99.2%. The lower Kaplan-Meier estimate at later time points reflects the progressively smaller number of knees remaining at risk and should be interpreted cautiously.

The Kaplan–Meier survival curve is shown in Figure 4, and corresponding survival estimates are summarized in Table 3.

Repeat CRFA

Repeat CRFA occurred in six patients, involving eight knees, because two patients underwent bilateral repeat procedures.

The median interval between the initial procedure and repeat CRFA was approximately 15 months (range, 12–20 months).

Table 3 Kaplan-Meier Repeat-CRFA-Free Survival

Time After CRFA	Number at Risk	Events	Repeat-CRFA-Free Survival	95% CI
12 months	451	1	99.8%	99.3–100.0
13 months	417	3	99.1%	98.1–100.0
17 months	275	1	98.7%	97.5–99.9
18 months	239	1	98.3%	96.9–99.7
20 months	146	2	96.9%	94.6–99.3

Table 4 Characteristics of Repeat CRFA Cases

Case	Initial Procedure	Repeat Procedure	Knee(s) Retreated	Time to Repeat CRFA
1	2024-06-11 left; 2024–06-18 right	2026-02-24	Bilateral	20 months
2	2024-06-12	2025-11-26	Right	17 months
3	2024-06-25	2025-12-03	Right	18 months
4	2024-09-09	2025-10-08	Bilateral	13 months
5	2025-04-07	2026-05-08	Left	13 months
6	2025-05-13	2026-05-20	Left	12 months

Details of the repeat procedures are summarized in Table 4.

At the approximately one-year telephone follow-up, 1 of 1,025 knees (0.1%) was reported to have been recommended and scheduled for total knee arthroplasty (TKA). The patient reported persistent inadequate improvement in knee pain rather than interval worsening. This planned TKA was reported as a separate secondary outcome and was not counted as an event in the repeat-CRFA-free survival analysis.

Complications

No major procedure-related complications were observed during the study period.

Two minor complications occurred: one superficial hematoma and one superficial skin burn. Both resolved completely with conservative treatment and did not result in permanent sequelae.

No procedure-related infection, vascular injury, persistent neurological deficit, or hospitalization was observed.

Overall, ultrasound-guided CRFA using the MDNT Technique demonstrated a favorable safety profile in this large consecutive cohort.

Discussion

The present study evaluated clinical outcomes after ultrasound-guided cooled radiofrequency ablation (CRFA) performed using the Matsuoka Direct Nerve Targeting (MDNT) Technique in a large consecutive cohort of 1,025 knees with symptomatic knee osteoarthritis (KOA). The principal observations were substantial early pain reduction, sustained pain relief among the 246 knees with available one-year NRS assessment, a low observed repeat-CRFA rate, and a favorable safety profile. These findings describe outcomes associated with CRFA performed using MDNT; they do not isolate an MDNT-specific treatment effect or establish superiority over conventional CRFA.

Several randomized controlled trials have demonstrated that genicular nerve CRFA provides clinically meaningful pain relief in patients with knee osteoarthritis. Davis et al first reported superior pain reduction following cooled radiofrequency ablation compared with intra-articular corticosteroid injection, while Chen et al subsequently demonstrated sustained clinical benefit for at least 12 months in multicenter randomized trials.[9–11] Our findings are consistent with these reports and extend the current evidence by demonstrating favorable outcomes in one of the largest consecutive single-center cohorts reported to date.

The technical feature emphasized in this study is direct nerve targeting. Ultrasound-guided genicular nerve procedures and anatomically refined targeting protocols have been described previously; therefore, MDNT should not be interpreted as the first ultrasound-guided approach. The proposed distinction is the standardized combination of direct sonographic visualization of a nerve-like target at the symptomatic site with concordant pain reproduction, sensory stimulation, and an immediate characteristic response to initiation of radiofrequency energy before definitive ablation. The present study evaluates this integrated workflow as a procedural modification rather than proving a new mechanism or superior targeting accuracy.

The MDNT Technique was developed to address this potential limitation by combining direct ultrasonographic visualization with physiological confirmation of the target nerve before definitive ablation. In the present technique,

localization was confirmed by four complementary findings: visualization of the target nerve, reproduction of the patient's characteristic pain during needle contact, concordant sensory stimulation, and the characteristic pain response immediately after initiation of radiofrequency energy. Rather than relying solely on anatomical estimation, this multimodal confirmation strategy was designed to increase confidence that the responsible sensory nerve had been accurately identified before ablation.

The favorable clinical outcomes observed in this cohort may reflect CRFA itself, patient selection, operator experience, direct target localization, or a combination of these factors. Because there was no comparator group, the study cannot determine whether MDNT improves efficacy, durability, or procedural precision relative to established landmark-based or other ultrasound-guided techniques. Accordingly, MDNT should be considered a hypothesis-generating technical framework requiring prospective comparative validation.

Another observation was the low repeat-CRFA rate: eight of 1,025 treated knees underwent repeat CRFA during the available observation period. Because patients entered the study continuously over two years, follow-up duration differed substantially. Kaplan-Meier analysis was therefore used to estimate freedom from repeat CRFA while accounting for censoring. Importantly, repeat CRFA is a treatment decision and not an objective definition of treatment failure; patients with recurrent symptoms could pursue other treatments, decline repeat treatment, or undergo arthroplasty. One knee (0.1%) was reported at approximately one-year follow-up to have been recommended and scheduled for TKA because of persistent inadequate pain improvement; this was reported separately and was not treated as a repeat-CRFA event. The survival estimates should therefore be interpreted specifically as repeat-CRFA-free survival.

The safety profile observed in this study was also favorable. No major procedure-related complications, including infection, permanent neurological deficit, vascular injury, or procedure-related hospitalization, were observed. Only two minor complications—a superficial hematoma and a superficial skin burn—occurred, and both resolved with conservative management without permanent sequelae. These findings are consistent with previous reports indicating that ultrasound-guided CRFA is a safe minimally invasive procedure when performed by experienced operators.[16]

This study has several strengths. It represents a large consecutive clinical cohort treated with a standardized protocol, provides nearly complete early NRS data, reports the entire cohort's observed repeat-CRFA events, and uses time-to-event analysis to account for unequal follow-up duration. The study also provides detailed procedural description that may facilitate future independent evaluation of the MDNT workflow.

Several limitations should be acknowledged. First, this was a retrospective single-center observational study without a concurrent comparator, so causal or superiority claims regarding MDNT cannot be made. Second, all procedures were performed by the technique developer, limiting assessment of reproducibility and creating potential operator- and intellectual bias. Third, the unit of analysis was the knee; bilateral knees from the same patient were included and within-patient clustering was not modeled. Fourth, one-year NRS data were available for 246 knees. This primarily reflects staggered enrollment and rolling one-year telephone follow-up, because many later-treated knees had not yet reached one year at manuscript preparation; however, some eligible patients could not be contacted, so nonresponse bias remains possible. The 246 knees should therefore be regarded as an available-case one-year cohort rather than a complete long-term cohort. Fifth, validated functional and quality-of-life outcomes such as WOMAC, KOOS, and EQ-5D were not collected. Finally, later Kaplan-Meier estimates were based on progressively smaller numbers at risk. These limitations underscore the need for prospective multicenter studies with independent operators, prespecified follow-up, patient-level clustering methods, functional outcomes, and an active comparator.

Taken together, these data provide hypothesis-generating real-world evidence on CRFA performed using the MDNT workflow. The findings support prospective multicenter comparative evaluation to determine whether the integrated direct-visualization and physiological-confirmation strategy offers measurable advantages over established targeting approaches.

Conclusion

Ultrasound-guided cooled radiofrequency ablation performed using the Matsuoka Direct Nerve Targeting (MDNT) Technique was associated with substantial early pain reduction, sustained pain relief among knees with available one-

year follow-up, a low observed repeat-CRFA rate, and high repeat-CRFA-free survival in this large consecutive cohort. Given the retrospective single-center, single-operator design and absence of a comparator group, these findings should be considered hypothesis-generating. Prospective multicenter comparative studies are needed to determine whether direct nerve targeting provides advantages over established CRFA targeting techniques.

Ethics Approval

This retrospective observational study was conducted at Matsuoka Emergency Clinic, Minamikyushu, Kagoshima, Japan.

The study protocol was approved by the Ethics Committee of Wide Area Medical Corporation EMS (Approval No. 2601; approved on June 1, 2024) before the initiation of the study. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Because of the retrospective nature of the study and the use of anonymized clinical data, the requirement for informed consent was waived by the Ethics Committee.

Disclosure

The MDNT Technique was developed by the senior author (Y.M.), who also performed all procedures in this study. This intellectual and operator involvement is acknowledged as a potential source of bias. The author declares no financial, consulting, research-funding, intellectual-property, patent, or other commercial relationships with Avanos Medical or any entity associated with the MDNT Technique. The author reports no other conflicts of interest in this work.

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